Machine Learning & Neural Networks Assignment

Topic: Logistic Regression & Backpropagation Gradients | **Student ID:** 2303031240866

Problem Statement:
Given the Binary Cross-Entropy (BCE) loss $\mathscr{L}(y,\hat{y}) = -[y\ln\hat{y} + (1-y)\ln(1-\hat{y})]$ where $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$, prove that the gradient with respect to logit z simplifies to $\frac{\partial\mathscr{L}}{\partial z} = \hat{y} - y$.

Student Solution Derivation:
Step 1: Derivative of the Sigmoid Activation Function
The sigmoid function is $\sigma(z) = (1+e^{-z})^{-1}$. Differentiating with respect to z:

$$\frac{d\hat{y}}{dz} = \frac{d\sigma(z)}{dz} = -(1+e^{-z})^{-2}(-e^{-z}) = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \hat{y}(1-\hat{y})$$

Step 2: Derivative of BCE Loss with respect to Prediction $\hat{y}$
Differentiating $\mathscr{L} = -y\ln\hat{y} - (1-y)\ln(1-\hat{y})$:

$$\frac{\partial\mathscr{L}}{\partial\hat{y}} = -\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}(-1) = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})} = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$$

Step 3: Chain Rule Application
Applying the chain rule $\frac{\partial\mathscr{L}}{\partial z} = \frac{\partial\mathscr{L}}{\partial\hat{y}} \cdot \frac{\partial\hat{y}}{\partial z}$:

$$\frac{\partial\mathscr{L}}{\partial z} = \left(\frac{\hat{y}-y}{\hat{y}(1-\hat{y})}\right) \cdot (\hat{y}(1-\hat{y})) = \hat{y} - y$$

Conclusion: The gradient simplifies directly to $\hat{y} - y$ (Prediction Residual). [Q.E.D.]